

Sphere Rigidity

MATHLIBANNEX / SELECTED READING SCOPE

The mathematical goal

If at least one of two real normed spaces is finite-dimensional, their unit spheres are isometric in the chord metric exactly when the ambient spaces are linearly isometric. Equivalently, a surjective unit-sphere isometry yields a linear isometry equivalence of the ambient spaces under that finite-dimensional hypothesis.

75 approved Cards in this selected reading scope. The catalog contains 188 distinct Cards across the four Projects.

Scope

Finite-dimensional real normed spaces and surjective isometries between their unit spheres, with the theorem stated when at least one ambient space is finite-dimensional. The formalization includes radial extension, dimension transfer, determinant and Pluecker methods, analytic orientation tools, finite recovery, and ball-volume rigidity used in the mathematical routes.

Scope limits

The ambient spaces are real, and at least one is finite-dimensional. The conclusion is the existence of a linear isometry equivalence. This result does not assert that a specified sphere isometry extends linearly. The selected Cards retain their recorded formal-source qualification and document-correspondence boundaries.

Mathematical routes

A surjective isometry between unit spheres first gives a radial map and transfers finite-dimensionality. Boundary-integral and orientation arguments then identify the Plücker bodies associated with the two norms. Determinant-maximizing frames, finite support selections and maximal-minor factorization provide linear recovery; a limiting and ball-volume argument yields an ambient linear isometry. This is an existence statement and does not claim that the prescribed sphere isometry extends linearly.

Each route shows its direct Cards and the reused prerequisites needed in the selected reading graph. Shared Cards count once within each route and once in the Catalog.

Radial extension and dimension transfer

A surjective isometry between unit spheres produces a radial map and transfers finite-dimensionality between the ambient spaces. The radial map has the recorded metric bounds; it is not asserted to be linear or globally isometric.

7 direct Cards + 0 reused prerequisites = 7 unique Cards. Shared Cards count once in this route.

- Key result: [Isometric unit spheres force equal finite dimensions](#)
- Key definition: [Radial extension of an isometry between unit spheres](#)

[Read this route with prerequisites](#)

Compact hulls and support-face selection

Compact hulls and support faces provide successive finite refinements and a common generator. The sets need not themselves be convex; nonemptiness and the stated finite-dimensional hypotheses remain in the individual Cards.

4 direct Cards + 0 reused prerequisites = 4 unique Cards. Shared Cards count once in this route.

- Key result: [A common generator selected from equal convex hulls](#)
- Key definition: [Successive maximum refinement by an ordered list of functionals](#)

[Read this route with prerequisites](#)

Determinants and maximal-minor factorization

Determinant differences and maximal minors turn coordinate changes into controlled algebraic data. Cramer coordinates and the uniqueness of a right factor will later translate matching minor vectors into a linear map.

7 direct Cards + 2 reused prerequisites = 9 unique Cards. Shared Cards count once in this route.

- Key result: [A unique right factor from proportional oriented maximal minors](#)
- Key result: [Cramer's rule for coordinates of a row](#)

[Read this route with prerequisites](#)

Piola identities and boundary integrals

Mollification, the Piola identity and determinant telescoping give integral identities for compact perturbations. Strong local convergence then passes to Lipschitz maximal minors, whose integrals are determined by boundary agreement.

10 direct Cards + 1 reused prerequisite = 11 unique Cards. Shared Cards count once in this route.

- Key result: [Boundary agreement determines maximal-minor integrals](#)
- Key result: [The divergence-free cofactor identity](#)

[Read this route with prerequisites](#)

Zero weak gradients and constancy

The weak equation makes local mollifications constant, giving local almost-everywhere constancy of the original function. Overlap and countable gluing arguments give global almost-everywhere constancy on an open preconnected domain. The separate continuous case upgrades this to pointwise constancy.

10 direct Cards + 0 reused prerequisites = 10 unique Cards. Shared Cards count once in this route.

- Key result: [Global almost-everywhere constancy on a preconnected domain](#)
- Continuous case: [A continuous function with zero weak gradient is constant](#)

[Read this route with prerequisites](#)

Orientation and signed Jacobian transfer

Metric inverse bounds and the absolute Jacobian formula provide signed transfer. A weak Piola identity shows that the transported sign has zero weak gradient; on a preconnected target the sign is almost everywhere constant and the signed integral is that sign times target volume.

7 direct Cards + 19 reused prerequisites = 26 unique Cards. Shared Cards count once in this route.

- Key result: [The signed Jacobian integral is a sign times target volume](#)
- Key result: [The transported Jacobian sign has zero weak gradient](#)

[Read this route with prerequisites](#)

Determinant-maximizing frames and finite recovery

An attained positive determinant maximum supplies well-conditioned dual frames. Weighted satellite arguments and a finite almost-norming family control the vectors needed for approximate linear recovery; the exact lower-unit condition remains explicit.

11 direct Cards + 1 reused prerequisite = 12 unique Cards. Shared Cards count once in this route.

- Key result: [A finite absolute-maximizing family almost norms the unit sphere](#)
- Key result: [A common inverse estimate from row Cramer](#)

[Read this route with prerequisites](#)

Absolute volume and ball rigidity

The absolute Jacobian integral of the radial map yields target-ball volume. A limiting linear contraction has the exact determinant-volume scaling between the two unit balls. Ball inclusion and strict compact-subset volume comparison give ball equality; the separate ball-equality result then gives norm preservation.

5 direct Cards + 28 reused prerequisites = 33 unique Cards. Shared Cards count once in this route.

- Key result: [The recovered contraction maps one unit ball onto the other](#)
- Key result: [Equality of seminorm balls under a linear bijection preserves the seminorms](#)

[Read this route with prerequisites](#)

Plücker bodies and linear isometry recovery

Boundary integrals and a common orientation compare the Plücker bodies of the two norms. Support slices, generator matching and maximal-minor factorization give approximate linear recovery, then the limiting volume argument gives an ambient linear isometry. This existence conclusion does not assert extension of a specified sphere isometry.

14 direct Cards + 57 reused prerequisites = 71 unique Cards. Shared Cards count once in this route.

- Key result: [The unit-sphere chord metric determines the ambient normed space](#)
- Key result: [All Plücker bodies determine the norm up to linear isometry](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#) [10](#) [11](#) [12](#)

Level 0 (21 Cards)

Level 0

Radial extension of an isometry between unit spheres

Extends a bijective sphere isometry to all vectors by retaining each radius and transporting its unit direction.

`MathlibAnnex.Sphere.radialExtension`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#) · [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#) · [The recovered contraction maps one unit ball onto the other](#) · [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Successive maximum refinement by an ordered list of functionals

Defines a nonempty compact set of survivors after maximizing finitely many functionals in order.

`MathlibAnnex.lexicographicRefine`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A support face is the convex hull of the maximizing generators

Identifies the points of a convex hull on a supporting level by discarding generators with zero weight.

`MathlibAnnex.convexHull_maxSlice_eq_supportFace`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

The convex hull of a compact set in a finite real coordinate space

Realizes the ordinary convex hull as a finite union of compact images of bounded-length convex combinations.

`MathlibAnnex.isCompact_convexHull_pi`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The average of contractive derivative generators lies in the body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

The family of maximal minors in increasing row order

Collects the determinants of all square submatrices with all columns and an increasing selection of rows.

`MathlibAnnex.Matrix.maximalMinors`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Maximal minors scaled by the reference ball volume](#) · [Right multiplication scales every maximal minor by one determinant](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#) · [Determinant integrals under strong \$L^n\$ convergence](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Cramer's rule for coordinates of a row

Expresses a row in the basis of the rows of an invertible square matrix through row-replacement determinants.

`MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unique right factor from proportional oriented maximal minors](#) · [Almost-contractive linear recovery with exact volume scaling](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Mollification by a normalized smooth kernel

Defines convolution with a nonnegative smooth kernel of integral one.

`MathlibAnnex.Mollification.mollify`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Strong local convergence of mollified derivatives](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A cofactor row by row replacement

Fixes the cofactor convention used in the divergence and determinant identities.

`MathlibAnnex.Piola.cofactorRow`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The divergence-free cofactor identity](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Zero integral of a compactly supported divergence

Makes all faces in a box divergence formula vanish by placing the support strictly inside the box.

`MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Gluing one almost-everywhere constant over a countable cover

Combines local exceptional sets using an inequality between restricted measures.

`MathlibAnnex.aeConstantOn_of_countableCover`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Divergence as the trace of a derivative

Defines divergence without choosing coordinates.

`MathlibAnnex.divergence`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Vanishing weak gradient tested by divergence](#) · [Weak Piola identity for a compactly supported test field](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Identifying constants on an open overlap

Uses positive measure to find a point where both almost-everywhere equalities hold.

`MathlibAnnex.aeConstants_eq_of_open_overlap`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Differentiating a local mollification through its kernel

Writes a directional derivative without differentiating the locally integrable input.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Inverse maps on open sets with global metric bounds

Records the maps, their inverse relations, and the quantitative bounds used in the orientation argument.

`MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A positive lower metric bound passes to the derivative](#) · [Weak Piola identity for a compactly supported test field](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A weighted determinant polynomial with satellites

Adds linear row-replacement contributions to a base determinant.

`MathlibAnnex.Satellite.satellitePolynomial`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Absolute maximization forces a near-maximal base](#) · [Each satellite norms its coefficient preimage in absolute value](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

An attained absolute determinant maximum

Selects a maximizing dual frame for a specified basis.

`MathlibAnnex.DeterminantFrame.exists_maximizingFrame`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Absolute maximization forces a near-maximal base](#) · [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A compact proper subset of a seminorm ball has smaller Haar measure

Finds a nonempty open part of the missing set and uses finiteness of the compact set to make the measure comparison strict.

`MathlibAnnex.SeminormBall.measure_lt`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The recovered contraction maps one unit ball onto the other](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Equality of seminorm balls under a linear bijection preserves the seminorms

Turns a ball-image equality into one seminorm inequality in each direction.

`MathlibAnnex.SeminormBall.map_eq`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A seminorm increment bound passes to every derivative direction

Takes a difference-quotient limit without imposing a positive lower bound.

`MathlibAnnex.FDeriv.norm_apply_le_seminorm_of_lipschitz`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [One boundary extension with all five analytic properties](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (20 Cards)

Level 1

Continuous linear maps bounded by a seminorm

Defines the pointwise contraction condition relative to a stored equivalent seminorm.

`MathlibAnnex.EquivalentSeminorm.IsContraction`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The radial extension is 3-Lipschitz

Separates radial variation from the change of unit direction to obtain the global bound with coefficient $1 + 2$.

`MathlibAnnex.Sphere.lipschitzWith_radialExtension`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#) · [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [A common orientation sign for the radial derivative average](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The inverse sphere isometry gives the inverse radial map

Shows that extending the inverse sphere isometry radially undoes the forward radial extension.

`MathlibAnnex.Sphere.radialExtension_leftInverse`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#) · [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [A common orientation sign for the radial derivative average](#)

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Level 1

A common generator selected from equal convex hulls

Uses a first maximizing slice and a separating sequence to obtain an original common point carrying a prescribed property.

`MathlibAnnex.exists_common_of_convexHull_eq`

Immediate Card prerequisites: [Successive maximum refinement by an ordered list of functionals](#) · [A support face is the convex hull of the maximizing generators](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 1

The satellite polynomial is linear in the maximal-minor vector

Expresses the configuration polynomial as a linear pairing with maximal minors.

`MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A weighted determinant polynomial with satellites](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Maximal minors scaled by the reference ball volume

Fixes the volume factor and increasing-row signs.

`MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#) · [The set average of derivative generators](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A unique right factor from proportional oriented maximal minors

Recovers a square factor and its determinant from proportional determinants of every ordered row tuple.

`MathlibAnnex.Matrix.existsUnique_factor_of_orientedMaximalMinorsProportional`

Immediate Card prerequisites: [Cramer's rule for coordinates of a row](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

Right multiplication scales every maximal minor by one determinant

Turns a square right factor into a common scalar on the family of maximal minors.

`MathlibAnnex.Matrix.maximalMinors_mul`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

Strong local convergence of mollified derivatives

Approximates the derivative in local L^n , where n is the dimension of the domain.

`MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub`

Immediate Card prerequisites: [Mollification by a normalized smooth kernel](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The divergence-free cofactor identity

Cancels the Hessian terms that arise when differentiating a cofactor row.

`MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`

Immediate Card prerequisites: [A cofactor row by row replacement](#)

Used by in this scope: [A component flux gives a determinant difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Determinant integrals under strong L^n convergence

Uses Hölder's inequality to turn strong L^n convergence of matrix fields into convergence of their determinant integrals.

`MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn`

Immediate Card prerequisites: [A determinant difference bound in the sup operator norm](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Vanishing weak gradient tested by divergence

Expresses a weak equation through compactly supported continuously differentiable vector fields.

`MathlibAnnex.WeakDivergenceZero`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#)

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#) · [The transported Jacobian sign has zero weak gradient](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

From local to global almost-everywhere constancy

Separates agreement of constants by connectedness from countable measure-theoretic gluing.

`MathlibAnnex.exists_aeConstantOn_of_local`

Immediate Card prerequisites: [Gluing one almost-everywhere constant over a countable cover](#) · [Identifying constants on an open overlap](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A positive lower metric bound passes to the derivative

Preserves the same lower constant in the difference-quotient limit.

`MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound`

Immediate Card prerequisites: [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [Signed transfer from the absolute Jacobian formula](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Absolute maximization forces a near-maximal base

A dominant determinant term prevents a large deficit in the base frame.

`MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [An attained absolute determinant maximum](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Positivity of the determinant maximum

A uniformly scaled dual basis supplies a positive comparison determinant.

`MathlibAnnex.DeterminantFrame.determinantMaximum_pos`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A strict lower bound on a norm's unit sphere

Defines the strict lower condition on a fixed norm unit sphere.

`MathlibAnnex.FiniteSup.FinMapAlmostIsometric`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The set of near-maximal dual frames

Records a determinant deficit while keeping all dual rows contractive.

`MathlibAnnex.DeterminantFrame.nearMaxFrames`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#) · [Each satellite norms its coefficient preimage in absolute value](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A nearby norming point gives an almost-norming evaluation

Two triangle inequalities transfer absolute norm attainment to a unit vector.

`MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The recovered contraction maps one unit ball onto the other

Upgrades ball inclusion to equality by strict compact-subset volume comparison.

`MathlibAnnex.PluckerRecovery.LimitCertificate.image_unitBall_eq`

Immediate Card prerequisites: [A compact proper subset of a seminorm ball has smaller Haar measure](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

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Level 2 (11 Cards)

Level 2

Isometric unit spheres force equal finite dimensions

Compares Hausdorff dimensions in both directions using the surjective Lipschitz radial maps.

`MathlibAnnex.Sphere.firank_eq`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Finite-dimensionality transfers across a sphere isometry

Uses the radial homeomorphism to transfer local compactness, then applies the finite-dimensionality criterion for real normed spaces.

`MathlibAnnex.Sphere.finiteDimensional_codomain`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A component flux gives a determinant difference

Expresses the change from one output-coordinate perturbation as a divergence.

`MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`

Immediate Card prerequisites: [The divergence-free cofactor identity](#)

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Zero weak gradient gives zero derivatives of local mollifications

Substitutes a translated smooth kernel into the weak equation and tracks the reflection sign.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Differentiating a local mollification through its kernel](#)

Used by in this scope: [Local almost-everywhere constancy from the weak equation](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Signed transfer from the absolute Jacobian formula

Transfers an integrable function without first assuming that the Jacobian sign is constant.

`MathlibAnnex.BilipschitzOrientation.signed_area_transfer`

Immediate Card prerequisites: [A positive lower metric bound passes to the derivative](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A common inverse estimate from row Cramer

Controls all near-maximal inverse frames by an explicit basis-dependent constant.

`MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`

Immediate Card prerequisites: [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Each satellite norms its coefficient preimage in absolute value

Varying one functional at an absolute maximizer forces endpoint attainment.

`MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [The set of near-maximal dual frames](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

The absolute Jacobian integral of a radial sphere extension

Identifies the integral of the absolute Jacobian over an open source unit ball with the volume of the closed target unit ball.

`MathlibAnnex.Sphere.radialExtension_integral_abs_det`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [A seminorm with two-sided bounds against a reference norm](#) · [A determinant difference bound in the sup operator norm](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 2

Compactness of the signed generator set

Passes compactness from contractions to their two signed minor images.

`MathlibAnnex.PluckerBody.isCompact_generators`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [The average of contractive derivative generators lies in the body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

The set average of derivative generators

Separates the volume-scaled integrand from the normalization of its average.

`MathlibAnnex.Plucker.derivativeAverage`

Immediate Card prerequisites: [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [The average of contractive derivative generators lies in the body](#) · [A common orientation sign for the radial derivative average](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

The symmetric convex body of contraction minors

Takes the real convex hull of both signs of every contraction generator.

`MathlibAnnex.PluckerBody.body`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [The average of contractive derivative generators lies in the body](#) · [Almost-contractive linear recovery with exact volume scaling](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (4 Cards)

Level 3

Smooth compact perturbations preserve the determinant integral difference

Builds a finite telescope from single-output perturbations with compactly supported fluxes.

`MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff`

Immediate Card prerequisites: [A component flux gives a determinant difference](#) · [Zero integral of a compactly supported divergence](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

Local almost-everywhere constancy from the weak equation

Turns constant mollifications into one almost-everywhere constant by choosing a common convergence point.

`MathlibAnnex.WeakDivergenceZero.nonempty_localAECOnstantAt`

Immediate Card prerequisites: [Zero weak gradient gives zero derivatives of local mollifications](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

One fixed satellite family almost norms every detected unit vector

Uses a detecting coefficient and its matching satellite without replacing the family.

`MathlibAnnex.Satellite.absoluteMaximizer_satellite_s_almost_norm`

Immediate Card prerequisites: [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 3

The average of contractive derivative generators lies in the body

Uses closed-convex average membership for the actual derivative-generator function.

`MathlibAnnex.Plucker.derivativeAverage_mem_body`

Immediate Card prerequisites: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#) · [The set average of derivative generators](#) · [The convex hull of a compact set in a finite real coordinate space](#)

Used by in this scope: [One boundary extension with all five analytic properties](#)

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Level 4 (5 Cards)

Level 4

Maximal-minor integral differences for Lipschitz perturbations

Passes the smooth compact-perturbation identity to Lipschitz maps through local strong convergence.

`MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith`

Immediate Card prerequisites: [Determinant integrals under strong \$L^n\$ convergence](#) · [Strong local convergence of mollified derivatives](#) · [Smooth compact perturbations preserve the determinant integral difference](#)

Used by in this scope: [Boundary agreement determines maximal-minor integrals](#) · [Weak Piola identity for a compactly supported test field](#)

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Level 4

Global almost-everywhere constancy on a preconnected domain

Combines local weak-gradient rigidity with topological and countable gluing.

`MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn`

Immediate Card prerequisites: [Local almost-everywhere constancy from the weak equation](#) · [From local to global almost-everywhere constancy](#)

Used by in this scope: [A continuous function with zero weak gradient is constant](#) · [Almost-everywhere constancy of the sign on a preconnected target](#)

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Level 4

A finite absolute-maximizing family almost norms the unit sphere

Chooses the inverse bound, net and weight before fixing one satellite family.

`MathlibAnnex.Satellite.exists_goodSatellitePackage`

Immediate Card prerequisites: [A common inverse estimate from row Cramer](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 4

One boundary extension with all five analytic properties

Keeps boundary agreement, contraction, derivative control, integrability and average membership on one witness.

`MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem`

Immediate Card prerequisites: [A seminorm increment bound passes to every derivative direction](#) · [The average of contractive derivative generators lies in the body](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

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Level 4

The chosen support slice consists of almost-isometric generators

Obtains one almost-isometric map from each point of the specified raw maximum slice.

`MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer_good`

Immediate Card prerequisites: [The satellite polynomial is linear in the maximal-minor vector](#) · [Compactness of the signed generator set](#) · [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 5 (5 Cards)

Level 5

Boundary agreement determines maximal-minor integrals

Turns pointwise equality on a seminorm sphere into a compactly supported Lipschitz perturbation.

`MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq`

Immediate Card prerequisites: [Maximal-minor integral differences for Lipschitz perturbations](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

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Level 5

A continuous function with zero weak gradient is constant

Removes the exceptional set using continuity and positivity of open sets.

`MathlibAnnex.WeakDivergenceZero.exists_eqOn`

Immediate Card prerequisites: [Global almost-everywhere constancy on a preconnected domain](#)

Used by in this scope: None in this selected scope

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Level 5

Weak Piola identity for a compactly supported test field

Uses coordinate perturbations to show that the signed pullback of a test-field divergence has zero integral.

`MathlibAnnex.BilipschitzOrientation.weak_piola`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#) · [Maximal-minor integral differences for Lipschitz perturbations](#) · [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

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Level 5

Almost-contractive linear recovery with exact volume scaling

Recovers one bijective linear factor with a norm estimate and exact volume scaling.

`MathlibAnnex.PluckerRecovery.nonempty_linearCertificate_of_pluckerBodies_eq`

Immediate Card prerequisites: [The chosen support slice consists of almost-isometric generators](#) · [The symmetric convex body of contraction minors](#) · [Cramer's rule for coordinates of a row](#) · [A finite absolute-maximizing family almost norms the unit sphere](#) · [A common generator selected from equal convex hulls](#)

Used by in this scope: [A volume-normalized linear contraction obtained by a limit](#)

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Level 5

Equal Plücker bodies yield matching generators with an almost-isometric representative

Selects common minor coordinates with a target almost-isometric representative.

```
MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_of_all_body_eq
```

Immediate Card prerequisites: [The chosen support slice consists of almost-isometric generators](#) · [The symmetric convex body of contraction minors](#) · [A finite absolute-maximizing family almost norms the unit sphere](#) · [A common generator selected from equal convex hulls](#)

Used by in this scope: None in this selected scope

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Level 6 (2 Cards)

Level 6

The transported Jacobian sign has zero weak gradient

Uses signed transfer and weak Piola with the same test field to obtain the weak equation on the target.

```
MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign
```

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Signed transfer from the absolute Jacobian formula](#) · [Weak Piola identity for a compactly supported test field](#)

Used by in this scope: [Almost-everywhere constancy of the sign on a preconnected target](#)

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Level 6

A volume-normalized linear contraction obtained by a limit

Takes a compact operator limit preserving the exact volume equation.

```
MathlibAnnex.PluckerRecovery.nonempty_limitCertificate_of_pluckerBodies_eq
```

Immediate Card prerequisites: [Almost-contractive linear recovery with exact volume scaling](#)

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

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Level 7 (2 Cards)

Level 7

Almost-everywhere constancy of the sign on a preconnected target

Obtains one constant from the weak equation and preconnectedness of the target.

`MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const`

Immediate Card prerequisites: [Global almost-everywhere constancy on a preconnected domain](#) · [The transported Jacobian sign has zero weak gradient](#)

Used by in this scope: [The signed Jacobian integral is a sign times target volume](#)

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Level 7

All Plücker bodies determine the norm up to linear isometry

Recovers a real linear equivalence preserving the two supplied norms.

`MathlibAnnex.EquivalentSeminorm.nonempty_linearIsometryEquiv_of_pluckerBodies_eq`

Immediate Card prerequisites: [Equality of seminorm balls under a linear bijection preserves the seminorms](#) · [A volume-normalized linear contraction obtained by a limit](#) · [The recovered contraction maps one unit ball onto the other](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

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Level 8 (1 Card)

Level 8

The signed Jacobian integral is a sign times target volume

Uses preconnectedness to obtain a constant, positive measure to identify its sign, and finite measure to integrate it.

`MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`

Immediate Card prerequisites: [Almost-everywhere constancy of the sign on a preconnected target](#)

Used by in this scope: [A common orientation sign for the radial derivative average](#)

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Level 9 (1 Card)

Level 9

A common orientation sign for the radial derivative average

Identifies the whole minor vector using one signed Jacobian integral.

`MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The set average of derivative generators](#) · [The signed Jacobian integral is a sign times target volume](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

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Level 10 (1 Card)

Level 10

A target contraction generator lies in the source body

Transfers membership through two extensions with the same boundary trace.

`MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry`

Immediate Card prerequisites: [One boundary extension with all five analytic properties](#) · [Boundary agreement determines maximal-minor integrals](#) · [A common orientation sign for the radial derivative average](#)

Used by in this scope: [Sphere isometries preserve the Plücker body](#)

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Level 11 (1 Card)

Level 11

Sphere isometries preserve the Plücker body

Combines generator transport and its inverse for every finite output dimension.

`MathlibAnnex.PluckerBody.eq_of_sphereIsometry`

Immediate Card prerequisites: [A target contraction generator lies in the source body](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

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Level 12 (1 Card)

Level 12

The unit-sphere chord metric determines the ambient normed space

Classifies existence of ambient linear isometries from the unit-sphere chord metric.

`MathlibAnnex.Sphere.nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_of_finiteDimensional`

Immediate Card prerequisites: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [All Plücker bodies determine the norm up to linear isometry](#) · [Sphere isometries preserve the Plücker body](#)

Used by in this scope: None in this selected scope

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